



Nkabom Collaborative

Smart Onion Disease Detection

In partnership with



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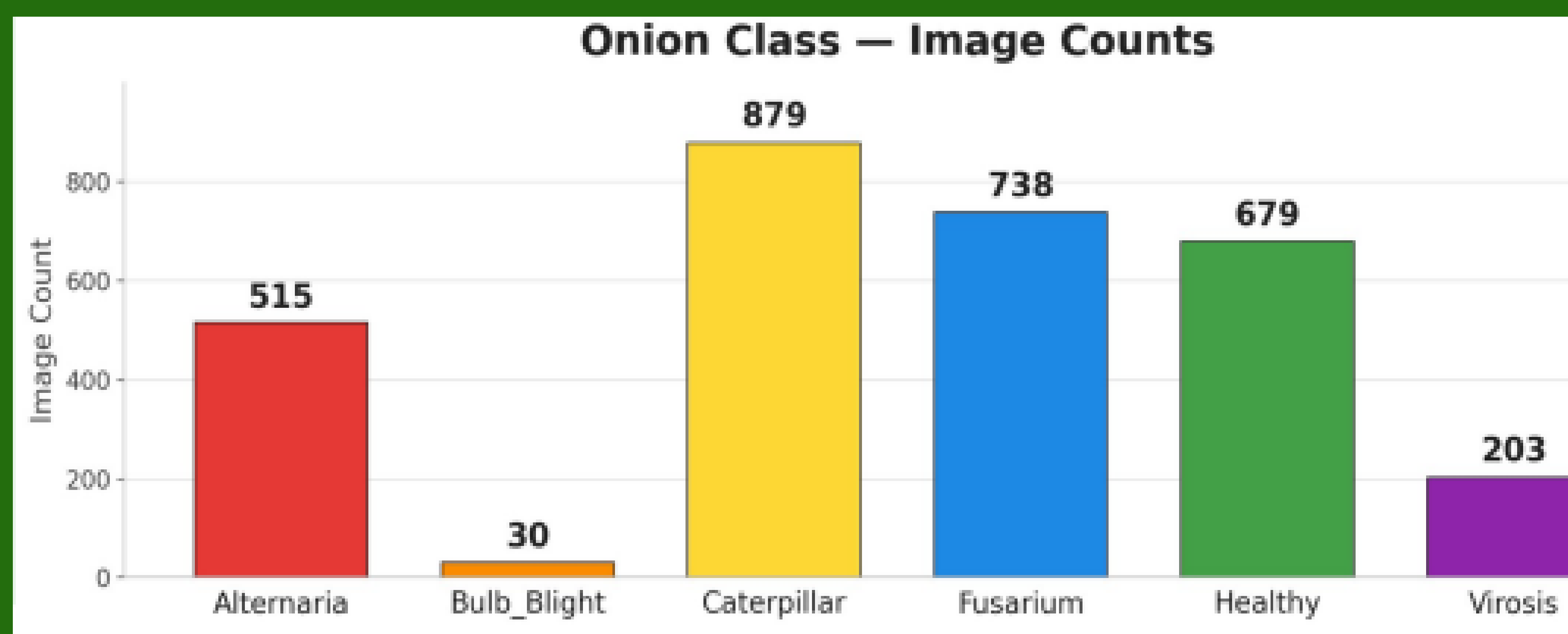
OnionGuard

1. The Problem

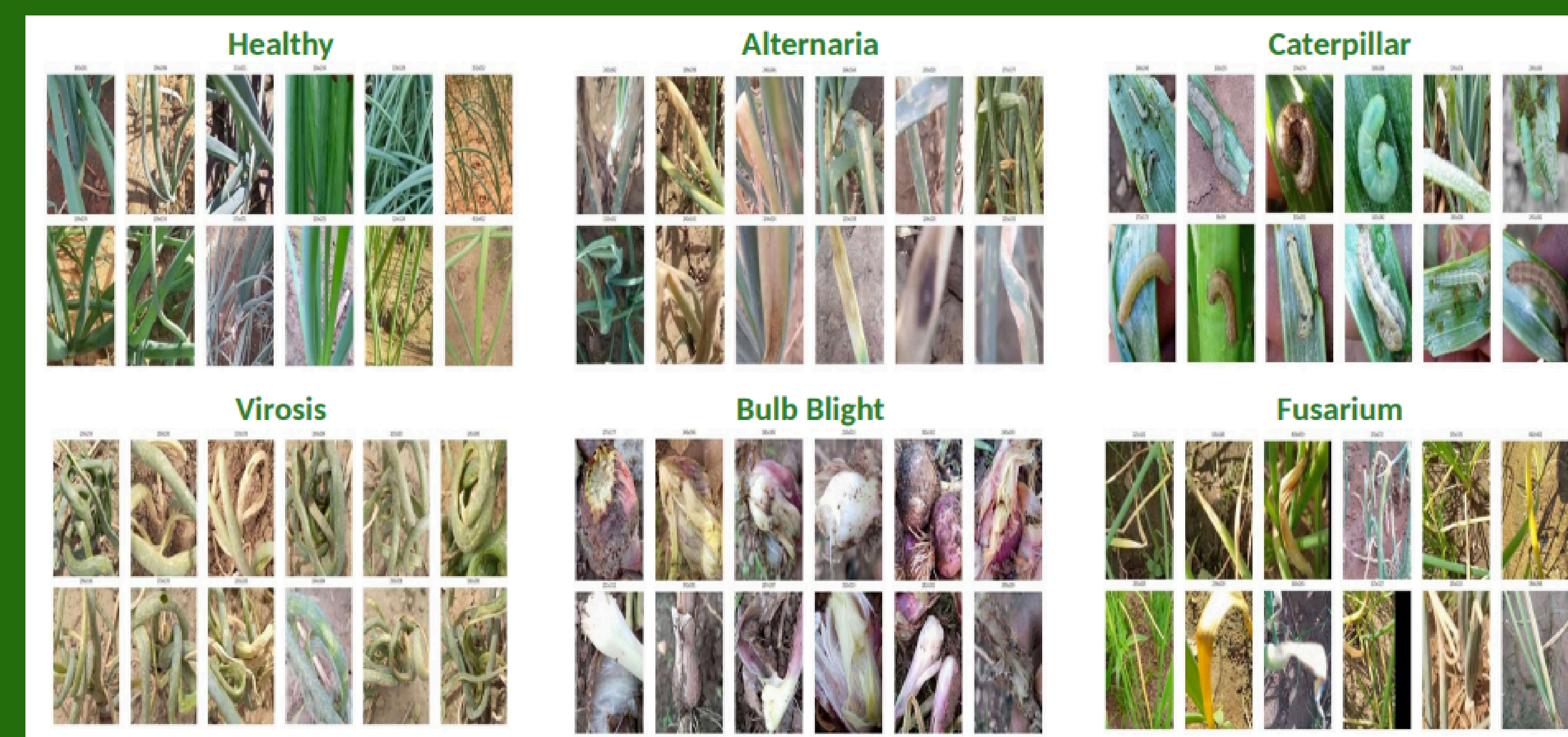
- 50% Crop losses in West Africa due to pests & diseases
- Poor rural extension access
- No affordable diagnostic tools for smallholder farmers
- No outbreak tracking system

2. Dataset

- Farm-collected dataset from Burkina Faso, data in Brief (Elsevier), 2025
- Covers healthy, pest, and disease classes
- First public onion benchmark for West Africa
- 3,044 images, 6 classes, resized to 224×224
- Imbalanced dataset handled with class weights & fine-tuning



3. Onion Disease Classes



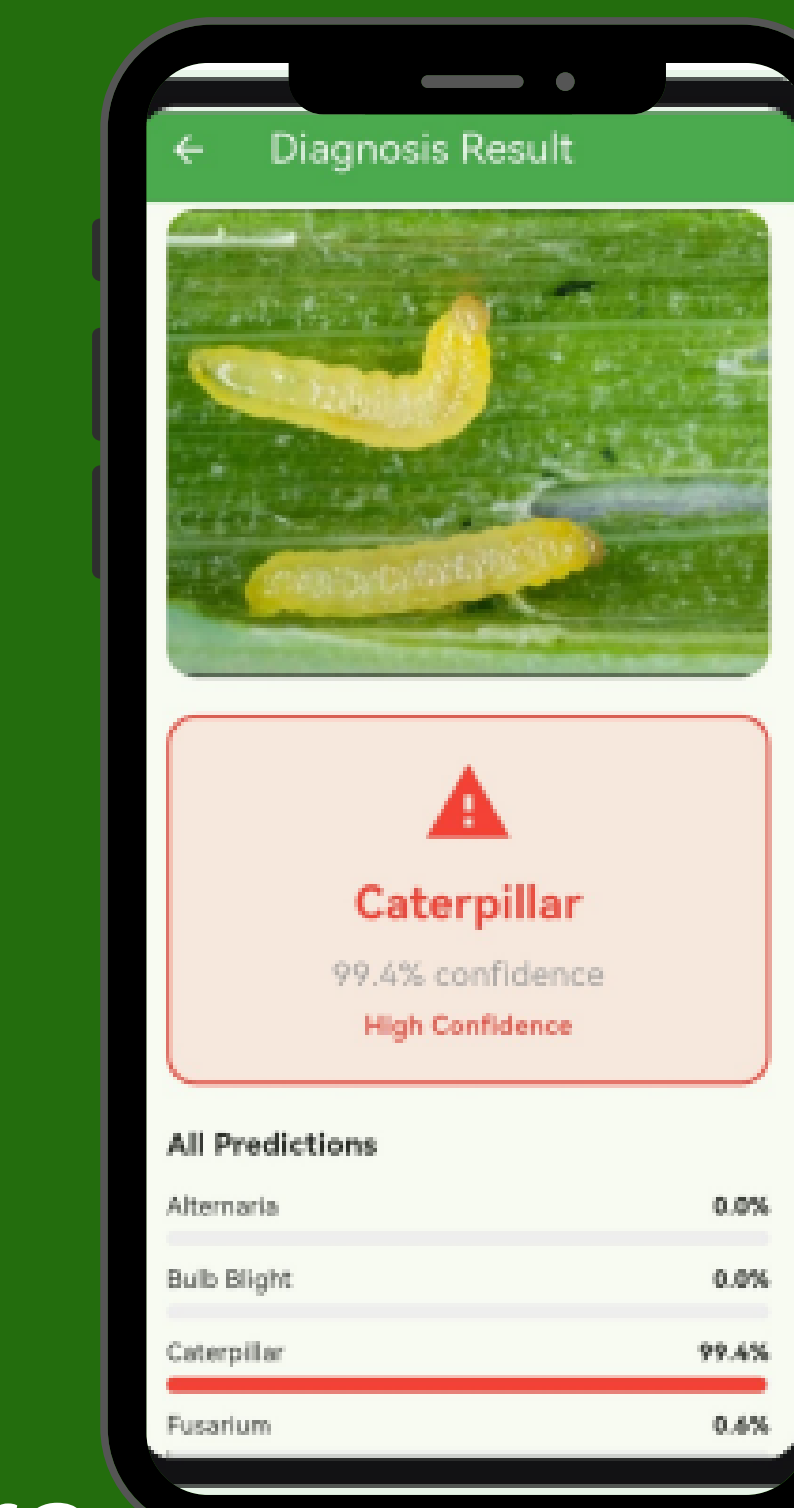
4. Our Solution

- On-device AI classifies 6 onion conditions
- Gemini AI for freshness detection
- Multilingual voice treatment guides
- Regional and personal analytics dashboards

Click here to download

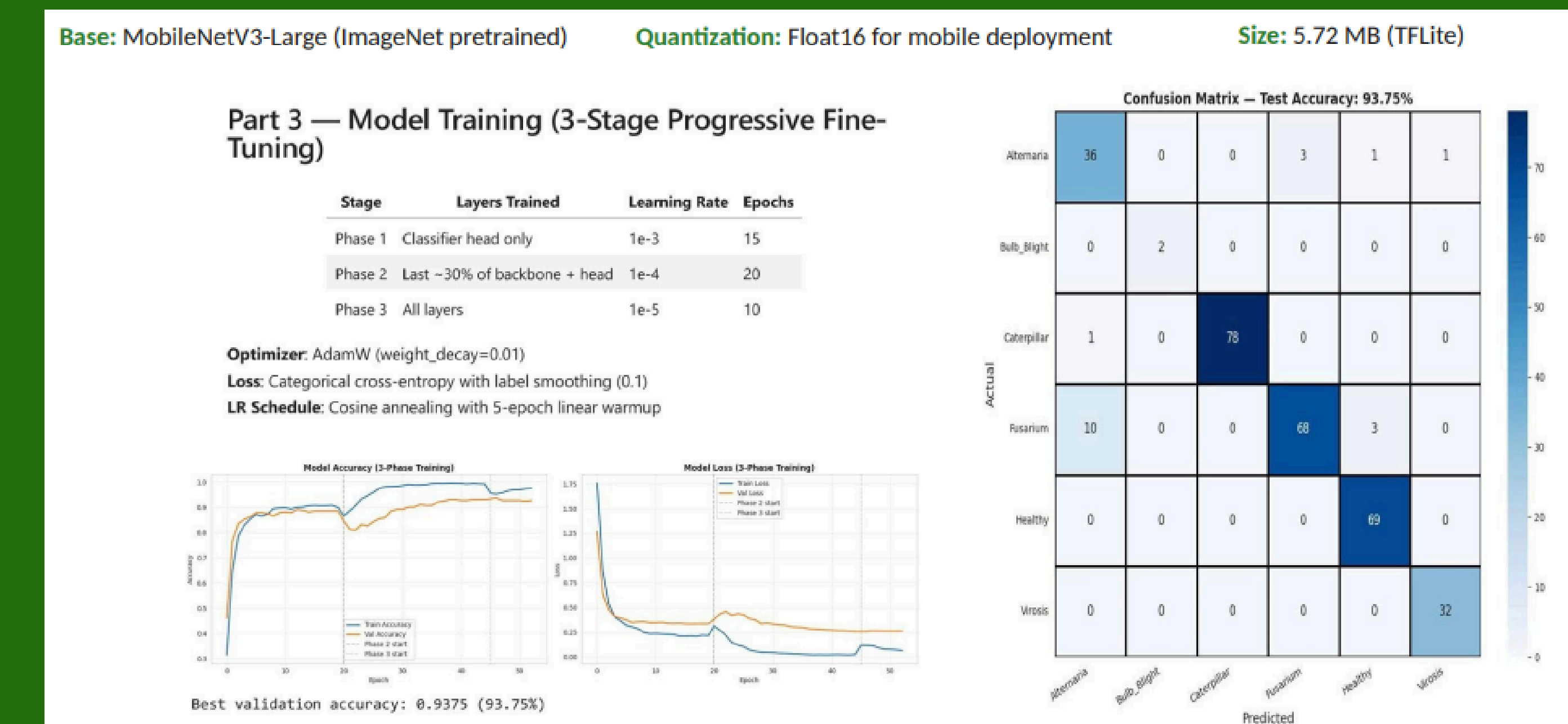


Live on Google Play Store



5. Methodology(ML Model Training)

93.75% Overall Test Accuracy



6. System Architecture

- Flutter App
- ↓
- Nginx + SSL
- ↓
- API Gateway (:8000)
- ↓
- Auth Service (:8001)
- Diagnosis Service (:8002)
- Treatment Service (:8003)
- Analytics Service (:8004)
- ↓
- MongoDB Atlas (Cloud DB)

7. Conclusion

- Impact: 9-language app cutting 50% onion crop loss.
- Tech: 93.75% accurate, 5.72MB on-device AI + Gemini Vision.
- Status: Live on Google Play; field-testing in Ghana; expanding to maize/tomatoes.

8. References

- Dataset • Appiah, M. K., et al. (2025)
- ML Model • Howard, A., et al. (2019)
- Google DeepMind. (2024).